Smart Tracking System using ESP32, BLE, and Blynk IoT Platform

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Abstract

In today's fast-paced world, people frequently misplace important personal belongings such as wallets, bags, keys, and luggage. To address this common issue, a smart tracking system was developed utilizing ESP32 microcontrollers, Bluetooth Low Energy (BLE), buzzer modules, push buttons, and the Blynk IoT platform. This paper presents the design and implementation of a two-part tracking system comprising a tracker device carried by the user and a tag device attached to the object. Structured as a Level 3 IoT system, the architecture integrates a comprehensive perception, processing, and application layer. The system continuously monitors the BLE Received Signal Strength Indicator (RSSI) to estimate distance and detect when an object is potentially lost. Upon detection, it automatically triggers mobile notifications, visualizes real-time RSSI vs. Time graphs on the cloud dashboard, and sounds audible alerts to assist in locating the misplaced item. Furthermore, this report analyzes the theoretical background of proximity detection and discusses the practical limitations of BLE tracking in real-world environments.

Index Terms

ESP32, BLE, IoT, Blynk, Smart Tracking, RSSI

I. Introduction

In today's world, people frequently misplace important belongings such as wallets, bags, keys, and luggage, leading to loss of valuable time and property. To solve this problem, a smart tracking system was developed to provide a reliable, low-cost, and user-friendly solution. The proposed system is built using ESP32 microcontrollers, Bluetooth Low Energy (BLE) technology, buzzer modules, push buttons, and the Blynk IoT platform.

The core of the system consists of two primary components: a Tracker Device carried by the user and a Tag Device attached to the object. The tracker continuously measures the BLE signal strength (RSSI) originating from the tag device. If the signal becomes weak for a certain duration, the system detects that the object may be moving far away. It then automatically sends a mobile notification through the Blynk application and rings a buzzer on the tag device, which helps the user locate the lost object.

II. Literature Review and Theoretical Background

A. Evolution of Tracking Technologies

The history of personal asset tracking has evolved significantly with the miniaturization of electronics and wireless communication. Traditional localization systems often relied on the Global Positioning System (GPS). While highly accurate outdoors, GPS suffers from high power consumption, bulky form factors, and poor indoor performance due to signal attenuation through buildings. To address indoor tracking, Radio Frequency Identification (RFID) systems have been widely deployed. RFID tags can be passive or active, but they generally require expensive reader equipment and lack seamless, direct integration with standard consumer mobile devices. Recently, Bluetooth Low Energy (BLE) has emerged as the preferred standard for proximity detection and indoor localization. Introduced as part of the Bluetooth 4.0 specification, BLE provides low power consumption, cost-effectiveness, and widespread compatibility with modern smartphones and microcontrollers. Unlike GPS or RFID, BLE allows for continuous tracking with minimal battery drain, making it ideal for tracking personal belongings like keys or wallets.

B. RSSI and Proximity Detection

Received Signal Strength Indicator (RSSI) is a measurement of the power present in a received radio signal. In BLE networks, RSSI is commonly used to estimate the distance between a transmitter (tag) and a receiver (tracker). Because radio signals attenuate as they travel through the air, the signal strength decreases logarithmically with distance. Unlike complex positioning algorithms such as Time of Flight (ToF) or Angle of Arrival (AoA), RSSI-based proximity detection is simple to implement and computationally lightweight, making it perfectly suited for resource-constrained microcontrollers like the ESP32. By continuously reading RSSI values, the system can categorize the tag's distance into generalized zones (e.g., Very Close, Nearby, Far). When the RSSI falls below a predefined threshold for a sustained period, the system detects an anomaly—indicating the object is moving out of the safe zone—and triggers a loss alert.

C. The Internet of Things (IoT) Paradigm

The Internet of Things (IoT) refers to the rapidly growing network of physical objects that feature IP connectivity, enabling communication between these devices, the cloud, and end-users. IoT extends internet connectivity beyond traditional computers to a diverse range of embedded devices. According to standard academic frameworks, IoT systems are classified into six hierarchical levels based on their complexity, processing location, network scale, and autonomy:

- **Level 1:** A single node that performs sensing/actuation, local analysis, and hosts the application locally.
- **Level 2:** A single node performs sensing, actuation, and local primary analysis. Data is stored in the cloud, and the application is cloud-based.
- **Level 3:** A single node sends raw data to the cloud. The cloud performs computationally intensive analysis and stores the data.
- **Level 4:** Multiple local nodes send data to the cloud for intensive analytics.
- **Level 5:** A large-scale wireless sensor network with multiple end nodes reporting to a local coordinator node.
- **Level 6:** Independent end nodes connect directly to the cloud with a centralized controller performing global optimization.

The integration of IoT into physical security and asset tracking allows for unprecedented situational awareness. By aggregating sensor data in the cloud, users can monitor their belongings in real-time.

III. System Architecture and IoT Level

This project is implemented as an **IoT Level 3** system. The Tracker ESP32 acts as the primary node that performs local sensing (scanning BLE RSSI) and forwards this data to the cloud (Blynk). The cloud platform hosts the application, stores the historical data, performs necessary analytics (such as generating RSSI vs. time graphs), and handles the intensive task of managing push notifications and remote monitoring. The tracker also handles actuation by triggering the buzzer on the tag when alerted.

The system architecture is divided into the hardware layer (Tracker and Tag) and the IoT/Cloud layer (Blynk Platform).

A. Tracker and Tag Devices

The system consists of:

- **A Tracker Device:** Carried by the user, this device acts as the central scanner and processing unit. It interfaces with the IoT platform to push data and receive manual commands.
- **A Tag Device:** Attached to the object of interest, this device acts as a BLE beacon, continuously advertising its presence.

B. Layered IoT Architecture

Following standard IoT design principles, the system is structured into five distinct operational layers:

- **Perception Layer:**
  - BLE Scanner (RSSI Signal Sensing)
- **Processing Layer:**
  - ESP32 Microcontrollers (Tracker and Tag)
- **Actuators (Local Alert):**
  - Buzzer Module (Audio)
- **Network Layer:**
  - Wi-Fi Network
  - HTTP/MQTT to Blynk Cloud
- **Application Layer:**
  - Blynk Cloud Database
  - Remote Monitoring Dashboard and Mobile App

IV. Hardware Design and Components

The hardware design focuses on portability and power efficiency. The key components include:

- **ESP32 Microcontrollers:** Serve as the core processing units for both the tracker and the tag, handling BLE communication and WiFi connectivity for IoT integration.
- **Buzzer Modules:** Integrated into the tag device to provide an audible alert when the object is lost or when a manual search is initiated.
- **Push Buttons:** Provided on the tracker device to allow the user to manually trigger the buzzer on the tag.

V. Software Methodology and Algorithms

The software logic is designed to continuously monitor proximity and determine the status of the tagged object through a series of steps.

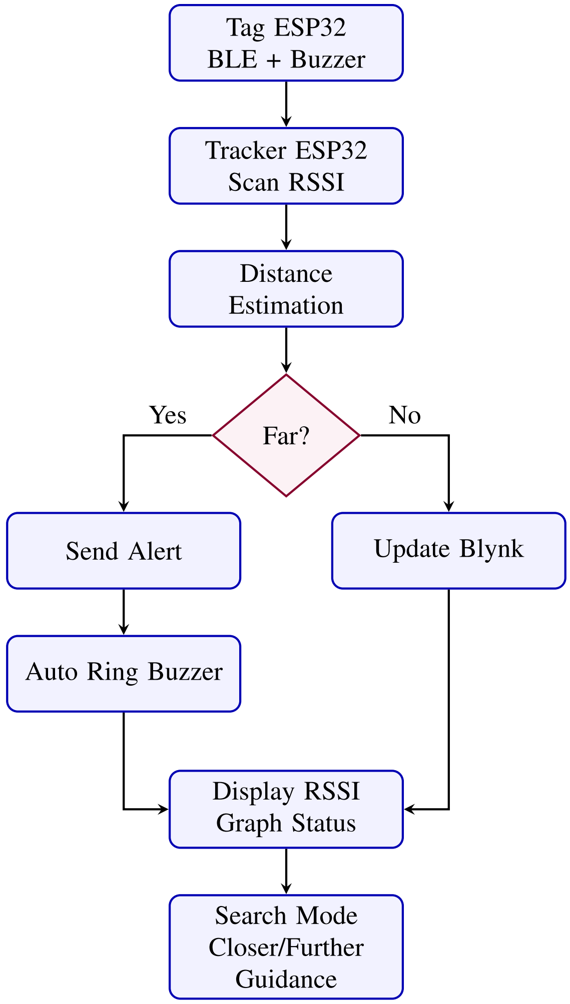

Fig. 1. Flowchart of the Smart Tracking System Algorithm.

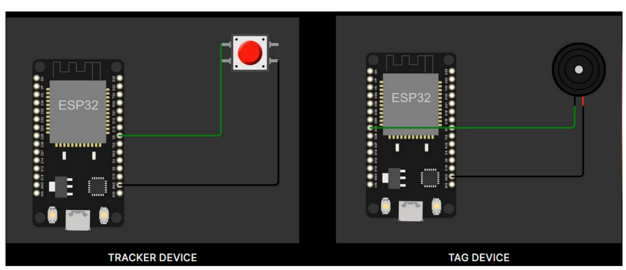

Fig. 2. Circuit Diagram of the Smart Tracking System.

A. Step 1: BLE Communication

The tag ESP32 continuously advertises BLE signals. Concurrently, the tracker ESP32 scans for the tag's specific BLE advertisement.

B. Step 2: Distance Estimation

The tracker reads the Received Signal Strength Indicator (RSSI) values from the tag. The RSSI value determines the estimated distance category of the object:

- **Very Close**
- **Nearby**

- **Far**

C. Step 3: Lost Detection

The system employs a counting mechanism to prevent false alarms caused by temporary signal fluctuations. If the RSSI remains continuously weak (indicating the object is ”Far”) and the Far Count reaches or exceeds 4 (Far Count ≥ 4), the system assumes the object may be lost.

D. Step 4: Automatic Alert

Upon detecting a lost object, the tracker executes the following automated actions:

- Sends a push notification to the user’s smartphone via the Blynk app.
- Sends a BLE command to the tag device to automatically activate its buzzer.

E. Step 5: Manual Search

The user can initiate a manual search by pressing a physical push button on the tracker device or via the app. This sends a command to the tag, causing the buzzer to ring and helping the user locate the object audibly.

F. Step 6: IoT Monitoring

The Blynk application provides a continuous monitoring interface that displays:

- Real-time RSSI values.
- Current distance status (Very Close, Nearby, Far).
- Lost alerts and notifications.
- An RSSI vs. Time graph to visualize signal trends and object movement.

VI. Results

The system was successfully implemented and tested in various indoor and outdoor environments. The distance estimation algorithm reliably categorized the proximity of the tag based on RSSI values.

A. RSSI Analysis and IoT Dashboard

Received Signal Strength Indicator (RSSI) is a critical metric in this system, measured in decibels relative to a milliwatt (dBm). It signifies the estimated power level of the Bluetooth signal received by the tracker from the tag. Because signal strength drops as the physical distance between the two devices increases, RSSI acts as a reliable proxy for proximity. A higher RSSI value (e.g., closer to 0, such as -40 dBm) indicates a strong signal and signifies that the object is ”Very Close” or ”Nearby”. Conversely, a lower RSSI value (e.g., -90 dBm) signifies a weak signal, implying the object is ”Far” or potentially lost.

The Blynk IoT dashboard provides a real-time visualization of these metrics. As shown in Figure 3, the interface displays the current distance category (e.g., ”Close”) and a live gauge of the exact RSSI value. Additionally, the dashboard features an **RSSI VS Time Graph**, which plots the signal strength over a selected time period (such as Live, 1h, 6h, or 1d). This graph is particularly useful for tracking the movement trend of the object. A sudden downward spike in the graph indicates the object is moving away rapidly, whereas a stable line indicates the object is stationary relative to the user.

During testing, when the tag was moved beyond the predefined safe range, the tracker consistently detected the drop in signal strength on the graph. After the Far Count threshold was met, the system successfully triggered the automatic buzzer on the tag and delivered push notifications to the Blynk app with minimal latency. The manual search functionality also proved highly effective in locating misplaced items within hearing range.

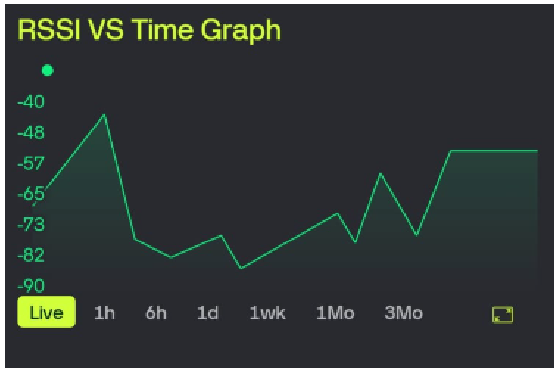

Fig. 3. Blynk IoT Dashboard showing Distance Category, live RSSI gauge, and the RSSI vs. Time Graph. (Replace 'dashboard_image.png' with your file)

VII. Limitations

While the smart tracking system provides a functional and cost-effective solution, it has several limitations inherent to the technologies used:

- **Limited BLE Range:** BLE signal strength naturally decreases over large distances, limiting the maximum tracking radius.
- **RSSI Fluctuation:** Physical obstacles, walls, and human bodies can significantly attenuate the signal, affecting RSSI accuracy and potentially causing false distance estimations.
- **Battery Consumption:** Continuous BLE scanning by the tracker device consumes substantial power, which may require frequent recharging.
- **No GPS Tracking:** The system relies purely on proximity and cannot provide exact geographic map coordinates if the object is lost outside the BLE range.
- **Indoor Signal Disturbance:** The 2.4 GHz frequency band used by BLE is shared with Wi-Fi and other wireless devices, which may cause signal interference and disturbance in busy indoor environments.

VIII. Future Works

Future iterations of this project could focus on the following enhancements:

- **Miniaturization:** Designing a custom Printed Circuit Board (PCB) to reduce the form factor of the tag, making it suitable for wallets and keys.
- **Power Optimization:** Implementing deep sleep modes and optimizing BLE advertising intervals to extend battery life significantly.
- **GPS Integration:** Incorporating a GPS module to record the last known location of the object when it disconnects from the BLE network.

IX. Conclusion

In conclusion, the developed smart tracking system effectively addresses the common problem of misplaced belongings. By leveraging the ESP32 microcontroller, BLE technology, and the Blynk IoT platform, the project provides a robust solution for proximity monitoring and alert generation. The combination of automatic lost detection, manual search capabilities, and real-time IoT monitoring ensures that users can easily keep track of their valuables and locate them quickly when misplaced.